

SEC P vs NP — Frameworks (live)

SEC (autonomous) — attributed to Ludovico Kubler

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Summary table

Framework ID	Status	Fitness	Generation	Parent	Name
fw_b9e7d103d0	ELABORATING	0.000	0	-	Ergodic Circuit Frame- work
fw_28b4bfb95f	ELABORATING	0.000	0	-	TROPICAL_FOURIER
fw_85a254b4a0	ELABORATING	0.000	0	-	Coarse Geomet- ric Karchmer- Wigderson (CG-KW)
fw_6997a27304	ELABORATING	0.000	0	-	Coarse Geomet- ric Lifting (CGL)

Details

Ergodic Circuit Framework (fw_b9e7d103d0)

- **Status:** ELABORATING
- **Fitness:** 0.000
- **Taxonomy:** COMM_COMPLEXITY
- **Target invariant:** communication_entropy_barrier (real number ≥ 0) $\rightarrow$ bounds The minimum communication complexity in a k-party number-on-forehead model, where lower bounds are derived from the growth rate of Kolmogorov-Sinai entropy in associated dynamical circuits. Aims to prove $\omega(\log n)$ lower bounds for explicit functions like Disjointness, avoiding natural proofs via non-uniform dynamics.

Primitives: - measurable_dynamical_circuit (tuple (C, X, μ , T)): A Boolean circuit C where gates are labeled by operations over a probability space (X, μ), and T: $X \rightarrow X$ is a measure-preserving transformation
- orbit_signature (function: $N \rightarrow [0,1]$): For a fixed input $x \in \{0,1\}^n$, the orbit signature is the sequence $(\mu(C(T^k(x))))_k$, capturing how the circuit's output evolves under

repeated applica - cross_correlation_flow (matrix $\in R^{\{d \times d\}}$): For a depth- d circuit, this matrix $F\{i,j\} = \int |C_i(x) - C_j(T^{(x)})|^2 d\mu(x)$ measures how perturbations via T propagate through layers. Computable vi

Tentative axioms: - A1: Any circuit with $o(n)$ communication complexity induces a dynamical system with sublinear Kolmogorov-Sinai entropy growth. - A2: Measure-preserving transformations corresponding to AC^0 circuits have bounded mixing time, while those for PSPACE-complete computations exhibit super-polynomial mixer-profile decay. - A3: If a family of circuits computes a function with high cross_correlation_flow norm, then its communication complexity is $\Omega(n^\delta)$ for some $\delta > 0$.

TROPICAL_FOURIER_ANALYSIS (fw_28b4bfb95f)

- **Status:** ELABORATING
- **Fitness:** 0.000
- **Taxonomy:** FOURIER_ANALYTIC
- **Target invariant:** MinimalFourierCoefficient (RealNumber) $\rightarrow$ bounds The discrepancy of a function under tropical Fourier analysis.

Primitives: - TropicalPolynomial (Function): A function defined over a tropical semiring (max-plus or min-plus) with computable coefficients. - TropicalFourierTransform (Transformation): A mapping from tropical polynomials to their Fourier coefficients over a tropical semiring. - DiscrepancyMeasure (Metric): A function quantifying the deviation of a tropical polynomial from uniformity.

Tentative axioms: - A1: TropicalConvolution preserves the tropical semiring structure under composition. - A2: TropicalFourierTransform is invertible when restricted to certain tropical polynomials. - A3: DiscrepancyMeasure is bounded by the maximum absolute value of Fourier coefficients.

Coarse Geometric Karchmer-Wigderson (CG-KW) (fw_85a254b4a0)

- **Status:** ELABORATING
- **Fitness:** 0.000
- **Taxonomy:** KARCHMER_WIGDERSON
- **Target invariant:** Coarse depth $\kappa(f)$ (non-negative real) $\rightarrow$ bounds $\kappa(f) := \sup$ over nontrivial $c \in HX^1(X_f)$ and $T \in C_u^*[X_f]$ of $\log|\langle c, T \rangle| / \log(\text{propagation}(T))$. The conjecture is $\text{depth}(f) \geq \kappa(f)$, so a super-logarithmic $\kappa(f)$ for an explicit f gives a super-logarithmic formula-depth lower bound (a $P \not\subseteq NC^1$ separation if κ is poly-large).

Primitives: - KW-metric space (finite metric space (X_f, d_f) with $X_f = f^{-1}(0) \sqcup f^{-1}(1)$): For a boolean function $f: \{0,1\}^n \rightarrow \{0,1\}$, define $d_f(x,y) = \log_2(\text{min number of coordinates a depth-bounded distinguisher must read to certify } f(x) \neq f(y))$. - Controlled cover (finite family $U = \{U_i \subset X_f\}$ with diameter bound R): A cover whose parts have d_f -diameter at most R and whose overlap multiplicity is at most m . Encoded as a hypergraph H_U (vertices = X_f , edges = U_i) - Coarse cocycle (function $c: X_f \times X_f \rightarrow \mathbb{Z}$ with finite support modulo controlled equivalence): A discrete 1-cochain in the Roe-style coarse cohomology $HX^1(X_f)$ restricted to the KW pair structure. Stored as a sparse table on pairs (x,y) with $d_f(x,y) \leq R$. - Asymptotic-dimension witness (vertex coloring $\chi: X_f \rightarrow [k]$ together with diameter bound R): Witnesses $\text{asdim}(X_f) \leq k-1$ in the sense of Gromov: each color class decomposes into d_f -bounded pieces of diameter $\leq R$ that are pairwise R -disjoint. - Roe-controlled operator (matrix $T \in R^{\{X_f \times X_f\}}$ with $T_{\{x,y\}} = 0$ whenever $d_f(x,y) > R$): Element of the algebraic uniform Roe algebra $C_u^*[X_f]$. Stored as a banded sparse matrix in the d_f -metric; multiplication, adjoint, and trace are al

Tentative axioms: - A1 (Coarse-KW link): formula depth $d(f) \geq \Omega(\log(\text{asdim}(X_f)))$ and more strongly $d(f) \geq \Omega(\kappa(f))$; proved by translating a KW protocol into a controlled cover whose multiplicity exponent equals depth. - A2 (Composition sub-additivity): $\text{asdim}(X_{\{f \circ g\}}) \geq \text{asdim}(X_f) + \text{asdim}(X_g) - O(1)$, giving a KRW-style direct-sum theorem at the level of coarse dimension. -

A3 (Anti-natural-proofs): for a uniformly random f , $HX^1(X_f)$ is generically zero (Roe-algebraic concentration of measure), so κ is NOT a ‘largeness’ property in the Razborov-Rudich sense — random f ’s - A4 (Anti-relativization): κ is a metric invariant of d_f and is destroyed by oracle access (oracle gates collapse d_f to ≤ 1), so κ -based bounds cannot relativize, in the spirit of Mendel-Naor metric - A5 (Roe-index rigidity): nontrivial classes in $HX^1(X_f)$ detected by the Roe-trace pairing are stable under coarse equivalence, mirroring Yu’s coarse Baum-Connes results (Yu 2000; Nowak-Yu 2012); expl

Coarse Geometric Lifting (CGL) (fw_6997a27304)

- **Status:** ELABORATING
- **Fitness:** 0.000
- **Taxonomy:** LIFTING
- **Target invariant:** CoarseLiftingComplexity (CLC) (N -valued function of $(f, G) \rightarrow$ bounds The deterministic 2-party communication complexity of $f \circ G^n$. Specifically: $CC(f \circ G^n) \geq Q(f) \cdot \log_2(\text{asdim}_R(G)+1)$ for the gadget scale $R = \text{diam}(G)$, with the goal of recovering and surpassing known query-to-communication lifting bounds (Raz-McKenzie, Göös-Pitassi-Watson) for gadgets where positive asymptotic dimension can be certified.

Primitives: - MetricGadget ($(X: \text{FiniteSet}, Y: \text{FiniteSet}, g: X \times Y \rightarrow \{0,1\}, d: (X \times Y)^2 \rightarrow \mathbb{R}_{\geq 0})$): A boolean gadget g equipped with a graph/Hamming metric d on $X \times Y$. Implementable as a tuple of two finite sets, a truth-table for g , and a precomputed - LiftedInputSpace $((X \times Y)^n, d_{\oplus})$: The n -fold product of a MetricGadget under the $\square^1$ (Hamming-sum) product metric $d_{\oplus}((u_i), (v_i)) = \sum d(u_i, v_i)$. Implementable as a generator over tup - RoeCover ($\text{List}[\text{Subset}((X \times Y)^n)]$ with multiplicity m and scale R): A finite cover of the lifted input space whose members have $d_{\oplus}$ -diameter $\leq R$ and any point is in $\leq m$ members. This is the discrete witness used in the - ProtocolPullback ($\text{Dict}[\text{transcript} \rightarrow \text{Subset}((X \times Y)^n)]$): The partition of inputs induced by transcripts of a (deterministic or randomized) communication protocol Π , viewed as a candidate RoeCover. Implementa - FolnerWitness ($\text{Sequence}[\text{Subset}((X \times Y)^n)]$): A nested family $F_1 \subset F_2 \subset \dots$ with $|\partial_R F_k|/|F_k| \rightarrow 0$ for each scale R , certifying amenability/Property A of the lifted space relative to a gadget.

Tentative axioms: - A1 (Coarse-Lift Functoriality): If two gadgets G_1, G_2 are coarsely equivalent (a bi-Lipschitz bijection up to bounded error), then $\text{CLC}(f, G_1) = \Theta(\text{CLC}(f, G_2))$ for every f . - A2 (Protocol $\rightarrow$ Cover): Every deterministic protocol Π for $f \circ G^n$ of cost c yields a RoeCover with multiplicity $\leq 2^c$ and bounded scale R_{Π} depending only on G ; hence $CC(f \circ G^n) \geq \log_2(\min \text{multiplicity})$ - A3 (Asdim Amplification): $\text{asdim}(G_1 \otimes G_2) \geq \text{asdim}(G_1) + \text{asdim}(G_2) - O(1)$, so iterated tensoring of a single ‘good’ gadget produces unbounded coarse dimension. - A4 (Property-A Transfer): If G has Property A and f has query complexity $Q(f)$, then $\text{CLC}(f, G) \geq Q(f) \cdot h(G)$ where $h(G)$ is the Hilbert-space compression exponent of G — a non-natural, non-relativizing - A5 (Non-Algebrization): The Roe algebra $C^*_R((X \times Y)^n)$ is not closed under polynomial extensions of the gadget alphabet; therefore lower bounds derived from its K -theory do not algebrize.